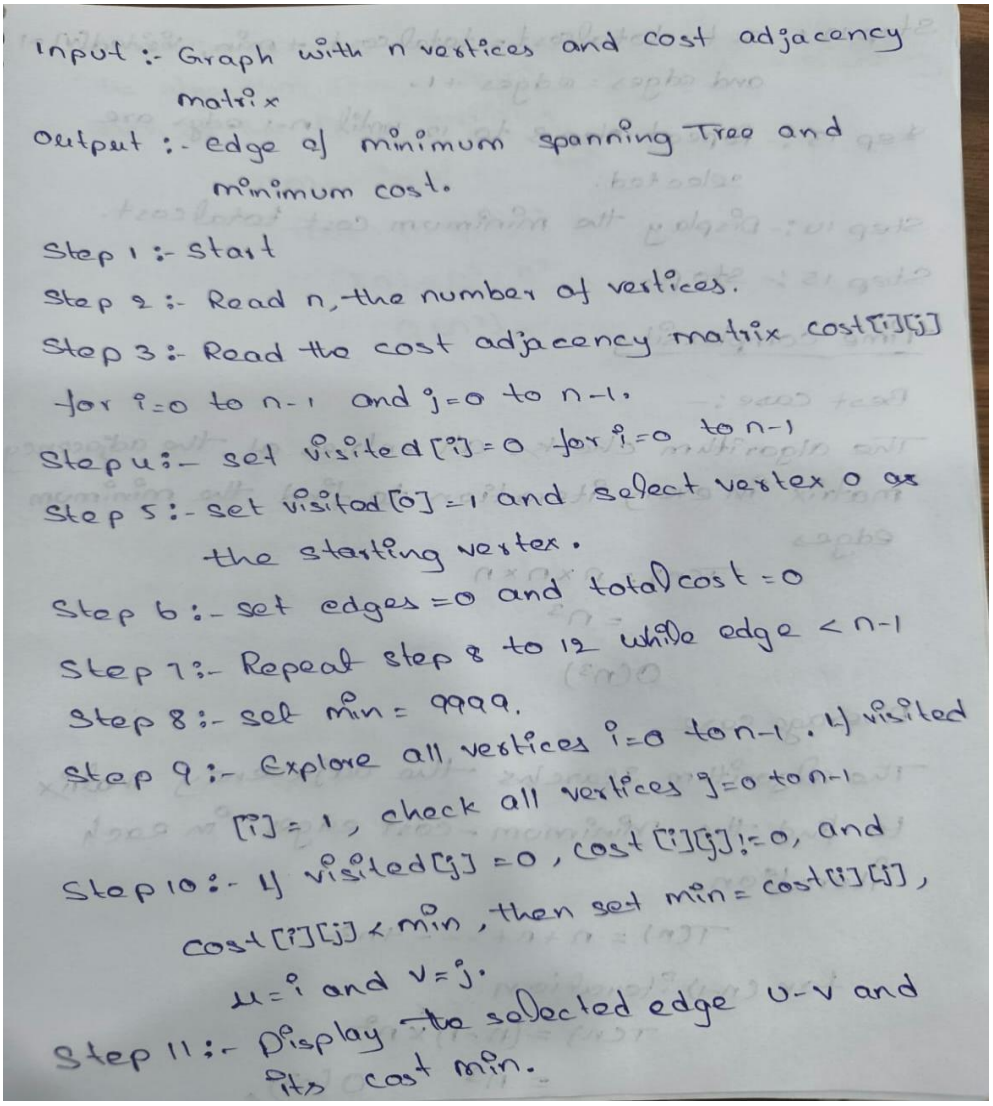
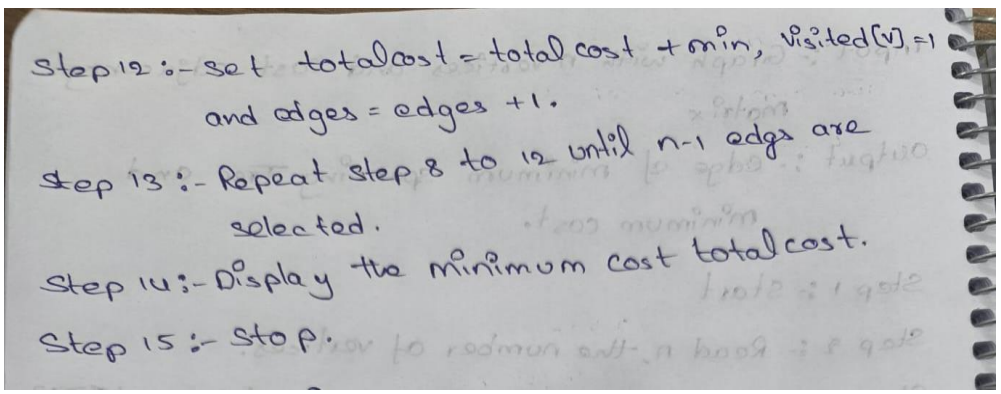
Practical- 9

Aim: To Implement prim's algorithm.

Prim's algorithm

Algorithm:





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Program:

```
#include <iostream>
using namespace std;

class Graph
{
public:
   int cost[10][10];
   int visited[10];
   int n;

   void getInput()
   {
      cout << "Enter number of vertices: ";
      cin >> n;

      cout << "Enter cost adjacency matrix:\n";

      for(int i = 0; i < n; i++)
      {
         for(int j = 0; j < n; j++)
         {
            cin >> cost[i][j];
         }
      }
   }

   void prim()
   {
      int edges = 0;
      int min;
      int u = 0, v = 0;
      int totalCost = 0;

      for(int i = 0; i < n; i++)
      {
         visited[i] = 0;
      }

      visited[0] = 1;

      cout << "\nEdges in Minimum Spanning Tree:\n";

      while(edges < n - 1)
      {
         min = 9999;

         for(int i = 0; i < n; i++)
         {
```

```
            if(visited[i] == 1)
            {
               for(int j = 0; j < n; j++)
               {
                  if(visited[j] == 0 && cost[i][j] != 0)
                  {
                     if(cost[i][j] < min)
                     {
                        min = cost[i][j];
                        u = i;
                        v = j;
                     }
                  }
               }
            }
         }

         cout << u << " - " << v << " = " << min << endl;

         totalCost = totalCost + min;

         visited[v] = 1;

         edges++;
      }

      cout << "\nMinimum Cost = " << totalCost << endl;
   }
};

int main()
{
   Graph g;

   g.getInput();

   g.prim();

   return 0;
}
```

Output:

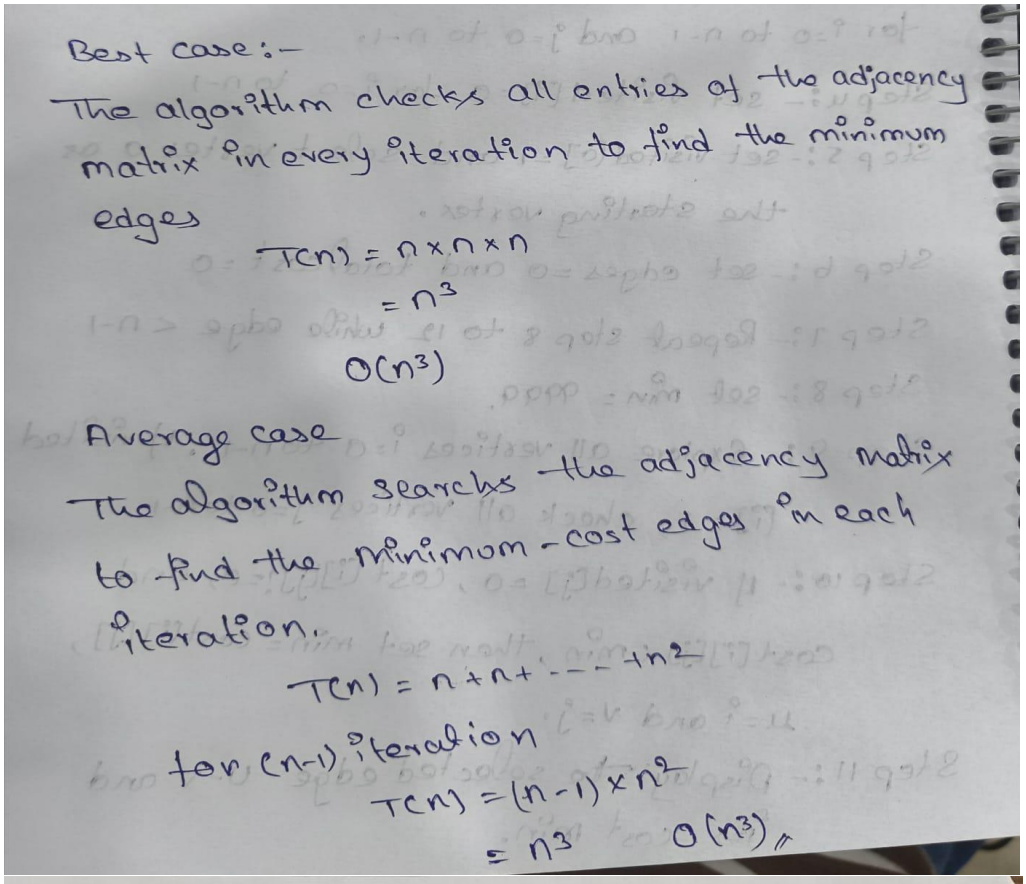
```
PS C:\MinGW\bin> & "C:\Users\chinn\Downloads\dsa in c++\practical-9\prim.exe"
Enter number of vertices: 6
Enter cost adjacency matrix:
0
2
0
6
0
0
2
0
3
8
5
0
0
3
0
0
7
0
6
8
0
0
7
0
6
8
0
0
9
4
0
5
7
9
4
1

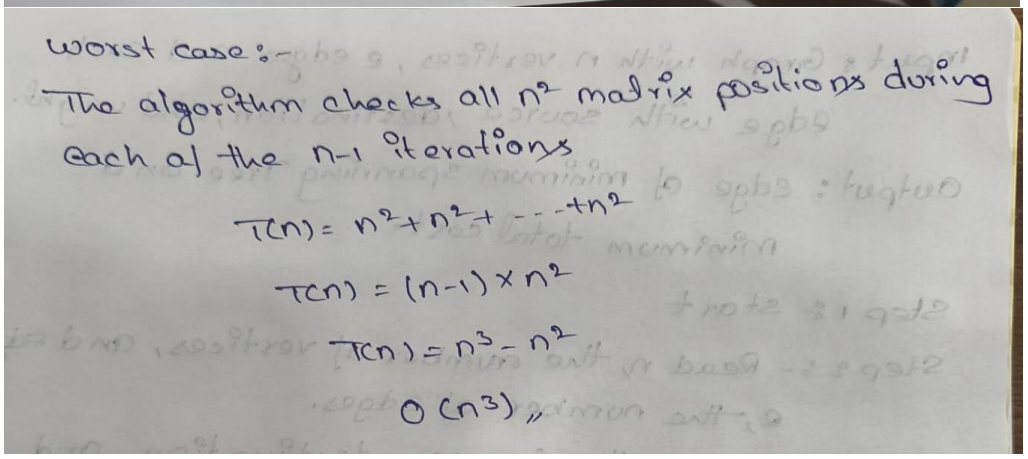
 Edges in Minimum Spanning Tree:
 0 - 1 = 2
 1 - 2 = 3
 1 - 4 = 5
 4 - 5 = 4
 0 - 3 = 6

 Minimum Cost = 20
```

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Time complexity:





Conclusion:

Thus, Prim's algorithm was successfully implemented to find the Minimum Spanning Tree (MST) of a connected, weighted graph. It selects the minimum-weight edges without forming any cycles and connects all vertices with the minimum possible total cost. This algorithm is useful in network design, road construction, and communication systems

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